

A Multistage Stochastic Transportation Model with Caputo-Type Congestion Memory, Demand Uncertainty, and Carbon Trading

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Abstract

This paper introduces a multistage stochastic transportation model that rigorously separates physical inventory conservation from hereditary operational persistence. Physical usable inventory is governed by a standard material-balance equation with proportional deterioration, while an auxiliary fractional congestion state—governed by a Caputo-type equation—captures persistent inbound-handling congestion. Demand uncertainty is represented by a finite scenario tree in which continuous sample paths are quantized into discrete nodal demand representatives, and nodal non-anticipativity constraints ensure that operational decisions depend only on information available at the corresponding decision epoch. The formulation incorporates explicit lost sales, transportation-mode activation, period-dependent carbon allowances, endogenous allowance trading, cumulative service-level guarantees, and a normalized convex combination of expected cost and Conditional Value-at-Risk.

To ensure mathematical well-posedness across the entire parameter domain, the Caputo operator is discretized via a piecewise L1 formula that explicitly defines the first-order limit without encountering undefined exponential expressions. The memory modulation constraint is formulated using only the beginning-of-period memory state and is linearized exactly into a pair of affine inequalities, preserving the mixed-integer linear structure. We prove positivity and decay of the discrete weights, convergence of the discrete fractional operator to the backward first-order difference as the fractional order approaches one, finite-horizon input-to-state boundedness of the discrete memory state, existence of an optimal policy under explicit boundedness assumptions, and the existence of an optimal nodal trading policy without simultaneous carbon-allowance buying and selling when the buying price strictly exceeds the selling price.

A Sample Average Approximation and Progressive Hedging framework is specified for computational implementation, with a probability-weighted scenario decomposition that correctly reproduces the normalized mean-CVaR objective. A supplementary computational archive accompanies the manuscript and contains the Python source code, generated scenario-tree instance, random seeds, raw Progressive Hedging histories, parameter-estimation profiles, and solver logs. A small synthetic study is reported as an implementation and reproducibility check; it is not intended as large-scale or empirical validation. Instead, it establishes a comprehensive reproducibility and validation protocol covering parameter identification, scenario-tree reduction error, deterministic-equivalent benchmarks, repeated-sample assessment, out-of-sample evaluation, and mixed-integer Progressive Hedging validation.

Keywords: fractional optimization, Caputo derivative, multistage stochastic programming, transportation planning, carbon trading, Conditional Value-at-Risk, Sample Average Approximation, Progressive Hedging, non-anticipativity.

1 Introduction

Transportation and distribution systems must simultaneously address economic efficiency, service continuity, uncertain demand, and environmental regulation. Standard multi-period transportation models usually represent operational states through first-order recursions. Such recursions are appropriate when the state at one period summarizes all relevant information from the past. They can, however, be restrictive when operational effects persist over several periods.

Examples of persistent effects include repeated handling losses, gradual deterioration, and long-lasting capacity degradation. Fractional calculus provides one possible phenomenological representation of such persistence (Caputo, 1967; Podlubny, 1999; Diethelm, 2010). In particular, a Caputo derivative depends on the complete history of state increments through a power-law kernel.

The use of a fractional operator does not, by itself, prove that a transportation system possesses fractional dynamics. Furthermore, a physical inventory state must rigorously obey a material conservation law. This paper therefore distinguishes physical usable inventory—which follows an ordinary conservation equation—from an auxiliary fractional state representing *persistent inbound-handling congestion*. Positive values of this memory state arise when delivered volume exceeds fulfilled demand, and they reduce future fulfillment capacity. Negative values represent handling slack or recovery capacity; they do not directly increase fulfillment beyond the physical-stock bound, but they can partially offset subsequent positive congestion through the memory dynamics.

Uncertainty is another essential component of transportation planning. Multistage stochastic programming provides a framework in which each decision depends only on observations available at its decision epoch (Birge and Louveaux, 2011; Shapiro et al., 2014). This information structure is more appropriate than a two-stage representation when shipments are revised sequentially and future demand is not revealed at the beginning of the planning horizon.

Environmental regulation is represented through period-dependent carbon allowances and allowance trading. Unlike a fixed emissions tax, a cap-and-trade mechanism permits the planner to purchase additional allowances or sell unused allowances. Related sustainable transportation and supply-chain models demonstrate the importance of jointly considering operational costs and emissions (Bektaş and Laporte, 2011; Asghari and Mirzapour Al-e hashem, 2021; Qu and Wu, 2021; Mirzaee et al., 2024; Gbadegoye et al., 2025).

Risk aversion is represented by Conditional Value-at-Risk (CVaR). CVaR is suitable for this purpose because it penalizes the severity of high-cost outcomes and possesses a tractable optimization representation (Rockafellar and Uryasev, 2000, 2002). We explicitly note that while the operational policy is multistage and non-anticipative, the risk measure is a static ex-ante CVaR evaluated over the total-horizon cost. Although this captures terminal tail risk without requiring nested conditional risk mappings, static ex-ante risk measures may exhibit temporal inconsistency across intermediate decision stages.

The contributions of this paper are as follows.

1. We formulate a multistage stochastic transportation model that rigorously separates physical usable inventory from an auxiliary fractional inbound-handling congestion state, incorporating explicit lost sales and cumulative service requirements.

2. We define a piecewise L1 discrete operator for the Caputo equation that remains well-defined over the full domain $\alpha \in (0, 1]$ and recovers the ordinary first-order difference at $\alpha = 1$.
3. We specify a scenario-tree construction procedure based on conditional distance clustering that quantizes continuous sample paths into discrete nodal demand representatives, preventing non-anticipativity from collapsing into artificial perfect information.
4. We derive a finite-dimensional deterministic equivalent under a normalized convex mean-CVaR objective, establish analytical properties including nodal carbon-trading separation, and prove finite-horizon input-to-state boundedness of the discrete memory state.
5. We specify a probability-weighted SAA-Progressive Hedging architecture that correctly decomposes the normalized mean-CVaR objective, without claiming global convergence for the mixed-integer case.
6. We provide a reproducible experimental protocol while excluding numerical results that are not supported by verifiable data, code, solver logs, and scenario-level outputs.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 introduces the fractional and stochastic concepts. Section 4.2 states the modeling and dimensional assumptions. Section 4.3 presents the multistage model. Section 5 establishes its analytical properties. Section 6 describes the computational methodology. Section 6.5 gives the validation protocol. Section 8.1 discusses managerial interpretation. Section 8.2 discusses limitations, and Section 10 concludes.

2 Related Literature

2.1 Stochastic transportation planning

Stochastic programming distinguishes decisions made before uncertainty is observed from decisions made after additional information becomes available. Two-stage models assume that the complete uncertain outcome is revealed between the first and second stages. Multistage models instead represent sequential information revelation (Birge and Louveaux, 2011; Shapiro et al., 2014).

This distinction is important in transportation planning. If shipments are selected period by period, a decision at period t should not depend on demands observed only after period t . The present paper therefore uses scenario histories and explicit non-anticipativity constraints.

Recent transportation models combine uncertain demand, capacity, emissions, and risk measures. For example, Gbadegoye et al. (2025) consider road-rail intermodal transportation under demand and capacity uncertainty. The present formulation differs by introducing an auxiliary fractional memory state and period-adapted carbon trading.

2.2 Sustainable transportation and carbon regulation

The pollution-routing problem demonstrates that transportation costs and emissions can lead to structurally different routing decisions (Bektaş and Laporte, 2011). The broader green vehicle-routing literature confirms that emissions depend on operational decisions and cannot always be treated as a fixed post-processing measure (Asghari and Mirzapour Al-e hashem, 2021).

Robust and stochastic supply-chain models have also incorporated carbon regulation and disruption risk (Qu and Wu, 2021; Mirzaee et al., 2024). The present paper treats allowance

purchases and sales as adaptive decisions. To avoid an unbounded trading representation, both variables have explicit finite bounds.

2.3 Fractional dynamics

Fractional derivatives provide a nonlocal representation of dynamics with hereditary effects (Caputo, 1967; Podlubny, 1999; Diethelm, 2010). The Caputo derivative is especially convenient when initial conditions are expressed using the state itself.

The L1 formula is a standard discretization of the Caputo derivative. Its coefficients are positive and decreasing, and it creates coupling between each period and all previous state increments (Lin and Xu, 2007). Faster approximations can become useful for long horizons (Zhang et al., 2022).

Fractional ideas have also appeared in inventory and supply-chain applications (Cuong et al., 2023; Haque et al., 2026). Nevertheless, the use of a fractional operator in optimization requires particular care. A physical inventory must respect mass conservation; fractional equations should therefore govern auxiliary memory or degradation states whose parameters must be calibrated jointly from empirical data.

2.4 Scenario-tree construction

Representing a continuous stochastic process by a finite scenario tree requires careful discretization to preserve the information structure. Dupačová et al. (2003) introduce scenario reduction via probability metrics, establishing that a reduced tree can approximate the original distribution. For multistage processes, tree quality must be evaluated using a filtration-aware metric such as the nested distance (Pflug and Pichler, 2012). Heitsch and Römisch (2009) extend this framework to multistage trees, providing algorithms for stagewise scenario reduction that preserve temporal dependence. These references motivate the scenario-tree construction protocol proposed for future implementation in this paper.

2.5 CVaR, SAA, and Progressive Hedging

The Rockafellar–Uryasev representation expresses CVaR through a scalar threshold and nonnegative excess variables (Rockafellar and Uryasev, 2000, 2002). This representation is compatible with finite-scenario stochastic optimization.

Sample Average Approximation replaces an expectation with an empirical average computed from sampled scenarios. Its reliability must be assessed through independent testing and repeated samples (Kleywegt et al., 2002; Shapiro et al., 2014). Randomized quasi-Monte Carlo sampling may also improve estimation in suitable risk-averse problems (Melnikov and Milz, 2025).

Progressive Hedging decomposes a scenario problem by temporarily relaxing non-anticipativity and penalizing disagreement between scenario copies (Rockafellar and Wets, 1991). Its classical convergence theory is based on convexity. In stochastic integer programming, it is commonly used as a heuristic, and its output must be validated against deterministic-equivalent solutions whenever possible (Christiansen et al., 2023).

3 Fractional and Stochastic Preliminaries

3.1 Caputo derivative

Definition 1 (Caputo derivative). *Let $w : [0, \tau_N] \rightarrow \mathbb{R}$ be absolutely continuous and let $0 < \alpha < 1$. The Caputo derivative of order α is*

$${}^C D_\tau^\alpha w(\tau) = \frac{1}{\Gamma(1-\alpha)} \int_0^\tau \frac{w'(s)}{(\tau-s)^\alpha} ds. \quad (1)$$

The kernel in Equation (1) gives nonzero weight to the full trajectory of past increments. The fractional order controls the decay pattern but should not be interpreted independently from the other dynamic parameters.

3.2 Normalized time and units

Let $t_{\text{ref}} > 0$ be a reference time and define

$$\tau = \frac{t_{\text{physical}}}{t_{\text{ref}}}.$$

The normalized time τ is dimensionless. Therefore, ${}^C D_\tau^\alpha w$ has the same physical unit as w .

If the fractional equation is written in physical time instead, then

$${}^C D_\tau^\alpha w = t_{\text{ref}}^\alpha {}^C D_{t_{\text{physical}}}^\alpha w.$$

The reference-time factor must consequently be retained when parameters are transferred between time scales.

3.3 Piecewise L1 discretization

Consider a uniform normalized grid $\tau_t = t\Delta\tau$ for $t = 0, \dots, N$. For $0 < \alpha < 1$, define the weights

$$a_\ell^{(\alpha)} = (\ell+1)^{1-\alpha} - \ell^{1-\alpha}, \quad \ell = 0, \dots, N-1. \quad (2)$$

To ensure that the discrete fractional operator remains formally defined over the entire domain $\alpha \in (0, 1]$ without generating undefined expressions of the form 0^0 at $\alpha = 1$, we define the piecewise discrete operator:

$$\mathcal{D}_{\Delta\tau}^\alpha w_t = \begin{cases} \frac{1}{\Gamma(2-\alpha)(\Delta\tau)^\alpha} \sum_{\ell=0}^{t-1} a_{t-\ell-1}^{(\alpha)} (w_{\ell+1} - w_\ell), & 0 < \alpha < 1, \\ \frac{w_t - w_{t-1}}{\Delta\tau}, & \alpha = 1. \end{cases} \quad (3)$$

Because $\Delta\tau$ is dimensionless, $\mathcal{D}_{\Delta\tau}^\alpha w_t$ has the same physical unit as w_t . When $\alpha = 1$, this operator directly recovers the backward first-order difference.

3.4 Scenario-tree construction via nodal quantization

When underlying demand uncertainty originates from continuous stochastic processes (such as drift-diffusion), continuous sample paths almost surely differ at every decision epoch ($d_{j1}^\omega \neq d_{j1}^{\omega'}$). Defining information equivalence by exact equality of continuous trajectories would collapse non-anticipativity into artificial perfect information, as every scenario would branch into its own node at period 1.

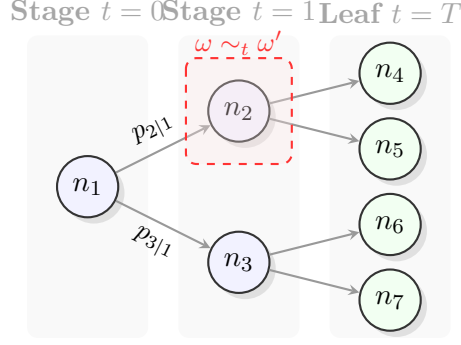


Figure 1: Stochastic information structure represented as a discrete scenario tree. Non-anticipativity conditions are enforced at nodes shared by multiple scenarios.

To prevent this, scenario histories must be constructed through a formal discretization and tree-reduction procedure (Dupačová et al., 2003; Heitsch and Römisch, 2009). The procedure operates as follows.

1. **Sample generation.** Generate a large ensemble of $M_0 \gg |\Omega|$ continuous demand trajectories from the underlying stochastic model.
2. **Stagewise conditional clustering.** At each period t , partition the trajectories passing through each parent node $n \in \mathcal{N}_{t-1}$ into $B_t(n)$ child clusters. Candidate reduced trees are evaluated using a filtration-aware metric, preferably nested distance.
3. **Nodal demand quantization.** Each cluster at period t defines a unique node $n \in \mathcal{N}_t$ with a representative demand vector $\hat{d}_{jtn}$. All scenarios passing through node n use this same representative demand at period t :

$$d_{jt}^\omega = \hat{d}_{jtn} \quad \text{whenever } n_t(\omega) = n. \quad (4)$$

Consequently, scenarios sharing a node are observationally indistinguishable, and their demand-balance and non-anticipativity constraints remain compatible.

4. **Conditional probabilities.** The probability of node n conditional on its parent is derived from cluster frequencies.
5. **Tree-reduction error.** Tree quality must be evaluated using a filtration-aware probability metric. Preferably, the nested distance between the original sampled process and the reduced scenario tree should be reported. When an ordinary Wasserstein metric is used as a computational approximation, this limitation must be stated explicitly.

Define $\Omega(n) = \{\omega \in \Omega : n_t(\omega) = n\}$. Let Ω be the resulting finite set of complete scenario paths with probabilities $p_\omega > 0$, where $\sum_{\omega \in \Omega} p_\omega = 1$. The information equivalence relation is defined nodally:

$$\omega \sim_t \omega' \iff n_t(\omega) = n_t(\omega'). \quad (5)$$

Decisions at period t must be identical for all scenarios sharing the same history node under $\sim_t$.

3.5 Conditional Value-at-Risk

For a finite-scenario cost C^ω , CVaR at confidence level $\beta \in (0, 1)$ is represented by

$$\text{CVaR}_\beta(C) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\beta} \sum_{\omega \in \Omega} p_\omega (C^\omega - \eta)_+ \right\}. \quad (6)$$

Introducing $\xi^\omega \geq 0$ gives the linear inequalities:

$$\xi^\omega \geq C^\omega - \eta, \quad \xi^\omega \geq 0. \quad (7)$$

4 Multistage Model

4.1 Ex-ante bounds on operational dynamics

To ensure computational tractability and numerical stability for the fractional memory state, we explicitly bound the realized deliveries and demand using:

$$\bar{d}_{jt} = \max_{n \in \mathcal{N}_t} \hat{d}_{jtn},$$

$$\bar{r}_{jt} = \min \left\{ \sum_{i \in I} S_{it}, \sum_{k \in K} Q_{kt} \right\}.$$

Consequently, the fractional forcing sequence is strictly bounded by:

$$|F_t^\omega| \leq \psi_j \left(\bar{r}_{jt} + \bar{d}_{jt} \right).$$

4.2 Modeling assumptions and dimensional conventions

Assumption 1 (State distinction). *The variable z_{jt}^ω denotes the physical usable inventory of customer j at the end of period t under scenario ω , governed by an ordinary material conservation law. A distinct state variable m_{jt}^ω represents an auxiliary Caputo-type congestion-memory state capturing persistent inbound-handling congestion generated when delivered volume exceeds fulfilled demand. Positive values of m_{jt}^ω reduce effective fulfillment capacity. Negative values of the memory state represent handling slack; they do not directly increase fulfillment but can partially offset subsequent congestion.*

Assumption 2 (Chronological sequence and demand timing). *Within each decision epoch t and history node $n_t(\omega)$, events follow a strict chronological order:*

- (i) observation of the nodal demand representative $\hat{d}_{jtn}$ defined in Equation (4);
- (ii) observation of beginning-of-period states $z_{j,t-1}^\omega$ and $m_{j,t-1}^\omega$;
- (iii) determination of operational delivery x_{ijkt}^ω , fulfilled demand q_{jt}^ω , and lost sales u_{jt}^ω ;
- (iv) evaluation of period carbon emissions E_t^ω ;
- (v) selection of allowance purchases b_t^ω and sales s_t^ω , modeled as an end-of-period compliance decision where realized operational emissions are known before trading;
- (vi) update of physical inventory z_{jt}^ω and memory state m_{jt}^ω .

Consequently, fulfillment decisions depend on $m_{j,t-1}^\omega$ (known at step (ii)), not on m_{jt}^ω (computed at step (vi)).

Assumption 3 (Negligible lead time). *Transportation lead times are assumed negligible relative to the length of one planning period. Hence x_{ijkt}^ω denotes both the quantity dispatched and delivered during period t , and supplier availability S_{it} applies at the same period.*

Assumption 4 (Lost sales). *Unfulfilled demand is lost rather than backlogged. A model with backlog would require a separate state variable and a different interperiod balance.*

Assumption 5 (Fractional interpretation). *The fractional equation governing m_{jt}^ω is a phenomenological constitutive model for inbound-handling congestion. The memory coupling coefficient ψ_j , fractional order α , and dissipation coefficient ρ_j must be estimated jointly from empirical observations. The operational interpretation is as follows: positive values of m_{jt}^ω represent persistent inbound-handling congestion generated when delivered volume exceeds fulfilled demand; negative values represent handling slack; they do not directly increase fulfillment beyond the physical inventory limit but can offset future congestion.*

Assumption 6 (Bounded planning domain). *Supply, transportation capacity, demand, usable inventory, memory states, carbon trading, and all cost coefficients are finite. The feasible set is assumed nonempty. Table 1 summarizes the dimensional conventions and physical units used throughout this paper.*

Table 1: Dimensional conventions and physical units.

Symbol	Interpretation	Unit	Comment
τ	normalized time	dimensionless	$\tau = t_{\text{physical}}/t_{\text{ref}}$
z	physical usable inventory	product	end-of-period physical stock
m	handling-congestion memory	product	L1-discretized persistent imbalance
x	dispatched and delivered quantity	product	negligible lead time assumed
q	fulfilled demand	product	quantity served within period τ_t
u	lost sales	product	unfulfilled part of demand
$\hat{d}$	nodal demand representative	product	$\hat{d} = q + u$ at each node
δ	inventory deterioration	dimensionless	physical stock proportional loss
ψ	memory coupling coefficient	dimensionless	scales the net-imbalance forcing
ρ	memory dissipation factor	dimensionless	estimated jointly with α
χ	memory modulation coefficient	dimensionless	reduces fulfillment when $m > 0$
e	emission factor	carbon/product	multiplied by delivered quantity
A, b, s	allowances and trades	carbon	period-dependent carbon quantities

4.3 Multistage stochastic transportation model

4.4 Sets and indices

- I : suppliers, indexed by i ;
- J : customers, indexed by j ;
- K : transportation modes, indexed by k ;
- $T = \{1, \dots, N\}$: planning periods, indexed by t ;

- Ω : scenario paths in the discrete tree, indexed by ω ;
- $\mathcal{N}_t$: set of history nodes at period t .

4.5 Parameters

- $p_\omega > 0$: probability of scenario ω ;
- $S_{it} \geq 0$: supply available from supplier i in period t ;
- $\hat{d}_{jtn} \geq 0$: nodal demand representative for customer j at node $n \in \mathcal{N}_t$;
- $c_{ijkt} \geq 0$: unit shipment cost;
- $f_k \geq 0$: fixed activation cost of mode k ;
- $Q_{kt} \geq 0$: period capacity of mode k ;
- $h_{jt} \geq 0$: physical inventory holding cost;
- $\pi_{jt} \geq 0$: lost-sales penalty;
- $e_{ijkt} \geq 0$: unit carbon emissions;
- $A_t \geq 0$: initial carbon allowance for period t ;
- $p_t^b > p_t^s \geq 0$: allowance buying and selling prices;
- $\bar{B}_t < \infty$: upper bound on allowance purchases;
- $\bar{S}_t < \infty$: upper bound on allowance sales;
- $z_j^0 \geq 0$: initial physical usable inventory;
- $\bar{z}_{jt} < \infty$: physical inventory capacity;
- $\delta_j \in [0, 1)$: physical inventory deterioration coefficient;
- $\psi_j > 0$: dimensionless memory coupling coefficient;
- $\rho_j \geq 0$: normalized memory dissipation coefficient;
- $\chi_j \in [0, 1)$: memory modulation coefficient on fulfillment. This memory modulation coefficient is calibrated separately from observations linking beginning-of-period congestion to reductions in effective fulfillment capacity;
- $\bar{m}_{jt} < \infty$: upper bound on absolute memory state;
- $\gamma_j \in [0, 1]$: minimum cumulative service fraction;
- $\alpha \in (0, 1]$: fractional order;
- $\beta \in (0, 1)$: CVaR confidence level;
- $\lambda \in [0, 1]$: normalized weight of the CVaR term;
- $\Delta\tau > 0$: normalized grid spacing.

4.6 Decision variables

The strategic decision is $y_k \in \{0, 1\}$, where $y_k = 1$ if mode k is activated. For each scenario ω and period t , define:

$x_{ijkt}^\omega \geq 0$	quantity dispatched and delivered from i to j via k ,
$q_{jt}^\omega \geq 0$	fulfilled demand,
$u_{jt}^\omega \geq 0$	lost sales,
$z_{jt}^\omega \geq 0$	physical usable inventory,
$m_{jt}^\omega \in \mathbb{R}$	inbound-handling congestion memory,
$b_t^\omega \geq 0$	allowances purchased,
$s_t^\omega \geq 0$	allowances sold.

The CVaR variables are $\eta \in \mathbb{R}$ and $\xi^\omega \geq 0$.

4.7 Scenario cost

The adaptive operating cost under scenario ω is

$$C^\omega = \sum_{t \in T} \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} c_{ijkt} x_{ijkt}^\omega + \sum_{t \in T} \sum_{j \in J} (h_{jt} z_{jt}^\omega + \pi_{jt} u_{jt}^\omega) + \sum_{t \in T} (p_t^b b_t^\omega - p_t^s s_t^\omega). \quad (8)$$

The fixed activation cost is kept outside C^ω because it is identical in every scenario.

4.8 Objective function

To avoid unnormalized double counting of operating costs and make the risk preference transparent, we adopt a normalized convex combination of expected cost and CVaR:

$$\min \sum_{k \in K} f_k y_k + (1 - \lambda) \sum_{\omega \in \Omega} p_\omega C^\omega + \lambda \left[\eta + \frac{1}{1 - \beta} \sum_{\omega \in \Omega} p_\omega \xi^\omega \right], \quad (9)$$

where $\lambda \in [0, 1]$. Here, $\lambda = 0$ represents pure risk-neutral expectation minimization, whereas $\lambda = 1$ represents pure CVaR minimization. Because the fixed activation cost is deterministic, translation equivariance implies that if total cost is $T = \sum_k f_k y_k + C$, then $(1 - \lambda)\mathbb{E}[T] + \lambda \text{CVaR}_\beta(T) = \sum_k f_k y_k + (1 - \lambda)\mathbb{E}[C] + \lambda \text{CVaR}_\beta(C)$, justifying the placement of the fixed cost outside the scenario-dependent term.

4.9 Supply and mode capacity

For all $i \in I$, $t \in T$, and $\omega \in \Omega$,

$$\sum_{j \in J} \sum_{k \in K} x_{ijkt}^\omega \leq S_{it}. \quad (10)$$

For all $k \in K$, $t \in T$, and $\omega \in \Omega$,

$$\sum_{i \in I} \sum_{j \in J} x_{ijkt}^\omega \leq Q_{kt} y_k. \quad (11)$$

4.10 Demand fulfillment, lost sales, and cumulative service

Because each scenario ω passing through node $n = n_t(\omega)$ uses the common representative demand $\hat{d}_{jtn}$, the demand split is written for all $j \in J$, $t \in T$, and $\omega \in \Omega$:

$$q_{jt}^\omega + u_{jt}^\omega = \hat{d}_{jt, n_t(\omega)}, \quad (12)$$

with

$$0 \leq q_{jt}^\omega \leq \hat{d}_{jt, n_t(\omega)}, \quad 0 \leq u_{jt}^\omega \leq \hat{d}_{jt, n_t(\omega)}. \quad (13)$$

To ensure that service continuity is maintained throughout the horizon rather than compensated only at the end, we impose a cumulative service requirement:

$$\sum_{\tau=1}^t q_{j\tau}^\omega \geq \gamma_j \sum_{\tau=1}^t \hat{d}_{j\tau, n_\tau(\omega)}, \quad j \in J, \quad t \in T, \quad \omega \in \Omega. \quad (14)$$

Equation (14) can make some scenarios infeasible; feasibility must therefore be checked for every generated instance.

4.11 Initial conditions

For every $j \in J$ and $\omega \in \Omega$,

$$z_{j0}^\omega = z_j^0, \quad m_{j0}^\omega = 0. \quad (15)$$

Initial values are common across scenarios because no uncertainty has been observed before the planning horizon.

4.12 Physical inventory balance and fractional memory state

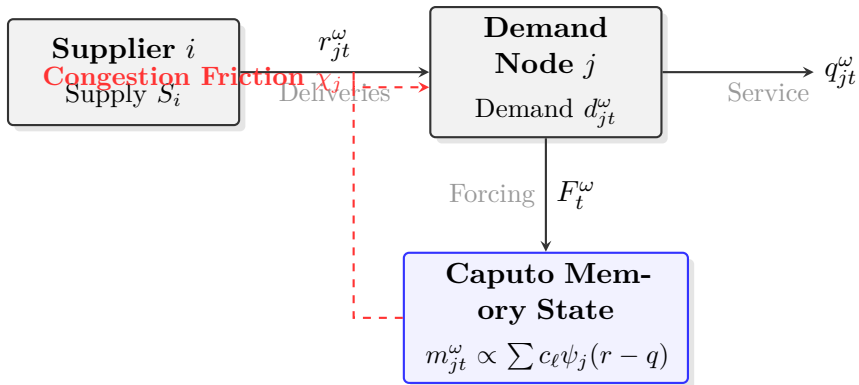


Figure 2: Coupled dynamics of physical inventory delivery and the fractional congestion memory state.

Let $r_{jt}^\omega = \sum_{i \in I} \sum_{k \in K} x_{ijkt}^\omega$ be the total quantity delivered to customer j during period t .

The physical usable inventory strictly respects material conservation with proportional deterioration $\delta_j \in [0, 1)$:

$$z_{jt}^\omega = (1 - \delta_j) z_{j, t-1}^\omega + r_{jt}^\omega - q_{jt}^\omega, \quad 0 \leq z_{jt}^\omega \leq \bar{z}_{jt}. \quad (16)$$

To capture persistent inbound-handling congestion, the auxiliary memory state m_{jt}^ω is governed by the piecewise discrete operator defined in Equation (3):

$$\mathcal{D}_{\Delta\tau}^\alpha m_{jt}^\omega = \psi_j \left(r_{jt}^\omega - q_{jt}^\omega \right) - \rho_j m_{jt}^\omega, \quad -\bar{m}_{jt} \leq m_{jt}^\omega \leq \bar{m}_{jt}. \quad (17)$$

The memory state modulates operational fulfillment capacity using the beginning-of-period memory $m_{j,t-1}^\omega$, which is known before any period- t decision is taken (see Assumption 2, step (ii)). This avoids instantaneous algebraic feedback between q_{jt}^ω and m_{jt}^ω . The resulting constraint is linearized exactly into the following pair of affine inequalities:

$$q_{jt}^\omega \leq (1 - \delta_j) z_{j,t-1}^\omega + r_{jt}^\omega, \quad (18a)$$

$$q_{jt}^\omega + \chi_j m_{j,t-1}^\omega \leq (1 - \delta_j) z_{j,t-1}^\omega + r_{jt}^\omega. \quad (18b)$$

Lemma 1 (Equivalence of memory modulation). *Constraints Equations (18a) and (18b) are jointly equivalent to the single nonlinear constraint*

$$q_{jt}^\omega \leq (1 - \delta_j) z_{j,t-1}^\omega + r_{jt}^\omega - \chi_j \max\{0, m_{j,t-1}^\omega\}.$$

Proof. Let $A_{jt}^\omega = (1 - \delta_j) z_{j,t-1}^\omega + r_{jt}^\omega$. If $m_{j,t-1}^\omega \leq 0$, then $\max\{0, m_{j,t-1}^\omega\} = 0$ and the nonlinear constraint reduces to $q_{jt}^\omega \leq A_{jt}^\omega$, which is exactly Equation (18a); meanwhile Equation (18b) reads $q_{jt}^\omega \leq A_{jt}^\omega - \chi_j m_{j,t-1}^\omega \geq A_{jt}^\omega$, so it is redundant. If $m_{j,t-1}^\omega > 0$, then $\max\{0, m_{j,t-1}^\omega\} = m_{j,t-1}^\omega$ and the nonlinear constraint becomes Equation (18b); now Equation (18a) is the weaker inequality. In both cases, the binding constraint from the pair matches the nonlinear expression. $\square$

Remark 1. *Because Equation (18) consists of two affine inequalities, the deterministic equivalent is a mixed-integer linear program for any fixed $\alpha \in (0, 1]$. In this dual-state architecture, Equation (16) ensures physical mass conservation, while Equation (17) acts as a phenomenological congestion filter. Because $r_{jt}^\omega - q_{jt}^\omega = z_{jt}^\omega - (1 - \delta_j) z_{j,t-1}^\omega$ by Equation (16), the memory state can also be interpreted as a fractional filter of sustained inventory accumulation. The model assumes that such accumulation generates persistent inbound-handling congestion. When $\alpha = 1$ and $\chi_j = 0$, the system reduces to a standard first-order inventory model without memory friction.*

4.13 Carbon accounting and trading

Period emissions are

$$E_t^\omega = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} e_{ijkt} x_{ijkt}^\omega. \quad (19)$$

Carbon allowances are period-specific: banking and borrowing across periods are excluded, and unused allowances can only be sold in the period in which they are allocated. Carbon feasibility requires

$$E_t^\omega \leq A_t + b_t^\omega - s_t^\omega, \quad t \in T, \omega \in \Omega. \quad (20)$$

Trading variables satisfy finite market bounds:

$$0 \leq b_t^\omega \leq \bar{B}_t, \quad 0 \leq s_t^\omega \leq \bar{S}_t. \quad (21)$$

4.14 CVaR constraints

For every $\omega \in \Omega$,

$$\xi^\omega \geq C^\omega - \eta, \quad \xi^\omega \geq 0. \quad (22)$$

Because all model variables are bounded, finite constants $\underline{C}$ and $\overline{C}$ can be constructed such that $\underline{C} \leq C^\omega \leq \overline{C}$. Without loss of optimality, we impose:

$$\underline{C} \leq \eta \leq \overline{C}, \quad 0 \leq \xi^\omega \leq \overline{C} - \underline{C}. \quad (23)$$

4.15 Non-anticipativity

For every pair $\omega, \omega' \in \Omega$ satisfying $\omega \sim_t \omega'$ as defined by nodal membership in Equation (5), impose:

$$x_{ijkt}^\omega = x_{ijkt}^{\omega'}, \quad i \in I, j \in J, k \in K, \quad (24a)$$

$$q_{jt}^\omega = q_{jt}^{\omega'}, \quad j \in J, \quad (24b)$$

$$u_{jt}^\omega = u_{jt}^{\omega'}, \quad j \in J, \quad (24c)$$

$$z_{jt}^\omega = z_{jt}^{\omega'}, \quad j \in J, \quad (24d)$$

$$m_{jt}^\omega = m_{jt}^{\omega'}, \quad j \in J, \quad (24e)$$

$$b_t^\omega = b_t^{\omega'}, \quad (24f)$$

$$s_t^\omega = s_t^{\omega'}. \quad (24g)$$

The variable η is common to all scenarios. The terminal excess variable ξ^ω remains scenario-dependent because it is evaluated after the complete trajectory has been observed.

Because Equation (4) assigns identical demand to all scenarios sharing a node, constraints Equations (12), (24b) and (24c) are mutually compatible.

5 Analytical Properties

5.1 Properties of the L1 coefficients

Lemma 2 (Positivity and decay). *For $0 < \alpha < 1$, the coefficients*

$$a_\ell^{(\alpha)} = (\ell + 1)^{1-\alpha} - \ell^{1-\alpha}$$

are strictly positive and nonincreasing in ℓ .

Proof. The function $\varphi(x) = x^{1-\alpha}$ is strictly increasing on $[0, \infty)$, so $a_\ell^{(\alpha)} = \varphi(\ell + 1) - \varphi(\ell) > 0$. Moreover,

$$\varphi''(x) = -\alpha(1-\alpha)x^{-1-\alpha} < 0 \quad (x > 0).$$

Thus φ is concave, and its forward differences are nonincreasing, yielding $a_{\ell+1}^{(\alpha)} \leq a_\ell^{(\alpha)}$. $\square$

5.2 First-order limiting case

Proposition 1 (Limit as $\alpha \rightarrow 1^-$). *For any fixed finite grid and fixed period t , the piecewise operator defined in Equation (3) satisfies:*

$$\lim_{\alpha \rightarrow 1^-} \mathcal{D}_{\Delta\tau}^\alpha w_t = \frac{w_t - w_{t-1}}{\Delta\tau}. \quad (25)$$

Proof. For $0 < \alpha < 1$, $a_0^{(\alpha)} = 1$, and for every integer $\ell \geq 1$,

$$\lim_{\alpha \rightarrow 1^-} a_\ell^{(\alpha)} = \lim_{\alpha \rightarrow 1^-} [(\ell + 1)^{1-\alpha} - \ell^{1-\alpha}] = 1 - 1 = 0.$$

Furthermore, $\Gamma(2 - \alpha) \rightarrow \Gamma(1) = 1$ and $(\Delta\tau)^\alpha \rightarrow \Delta\tau$. Only the term corresponding to $\ell = 0$ remains in the summation of Equation (3), giving $(w_t - w_{t-1})/\Delta\tau$, which matches exactly the definition of $\mathcal{D}_{\Delta\tau}^1 w_t$. $\square$

5.3 Finite-horizon boundedness of the memory state

Lemma 3 (Finite-horizon input-to-state bound). *For any fixed operational trajectory $\{r_{j\tau}^\omega, q_{j\tau}^\omega\}_{\tau=1}^t$ with $m_{j0}^\omega = 0$ and $\rho_j \geq 0$, the discrete memory state satisfying Equation (17) obeys*

$$|m_{jt}^\omega| \leq K_t^{(\alpha)} \max_{1 \leq \tau \leq t} |F_\tau^\omega|, \quad (26)$$

where $F_\tau^\omega = \psi_j(r_{j\tau}^\omega - q_{j\tau}^\omega)$ and the constants $K_t^{(\alpha)}$ depend only on $t, \alpha, \Delta\tau, \rho_j$. In particular, $K_t^{(\alpha)}$ is finite for every finite t but depends on the horizon.

Proof. Define $g_\alpha = \Gamma(2 - \alpha)(\Delta\tau)^\alpha$ and, for $s \geq 1$,

$$b_s^{(\alpha)} = a_{s-1}^{(\alpha)} - a_s^{(\alpha)} \geq 0,$$

where nonnegativity follows from the monotone-decay property of the L1 weights (Lemma 2). Using $m_{j0}^\omega = 0$ and rearranging the L1 discretization of Equation (17), one obtains for $0 < \alpha < 1$:

$$(1 + g_\alpha \rho_j) m_{jt}^\omega = g_\alpha F_t^\omega + \sum_{\ell=1}^{t-1} b_{t-\ell}^{(\alpha)} m_{j\ell}^\omega. \quad (27)$$

Define recursively:

$$K_0^{(\alpha)} = 0, \quad K_t^{(\alpha)} = \frac{g_\alpha + \sum_{\ell=1}^{t-1} b_{t-\ell}^{(\alpha)} K_\ell^{(\alpha)}}{1 + g_\alpha \rho_j}.$$

We prove Equation (26) by induction. The base case $|m_{j0}^\omega| = 0 \leq K_0^{(\alpha)} \max_\tau |F_\tau^\omega|$ holds. Suppose $|m_{j\ell}^\omega| \leq K_\ell^{(\alpha)} \max_\tau |F_\tau^\omega|$ for all $\ell < t$. From Equation (27), taking absolute values and using $b_s^{(\alpha)} \geq 0$:

$$(1 + g_\alpha \rho_j) |m_{jt}^\omega| \leq g_\alpha |F_t^\omega| + \sum_{\ell=1}^{t-1} b_{t-\ell}^{(\alpha)} |m_{j\ell}^\omega| \leq \left(g_\alpha + \sum_{\ell=1}^{t-1} b_{t-\ell}^{(\alpha)} K_\ell^{(\alpha)} \right) \max_\tau |F_\tau^\omega|,$$

which gives $|m_{jt}^\omega| \leq K_t^{(\alpha)} \max_\tau |F_\tau^\omega|$. Because the horizon is finite and all coefficients $b_s^{(\alpha)}$ are nonnegative and finite, each $K_t^{(\alpha)}$ is finite.

When $\alpha = 1$, the memory equation reduces to $m_{jt}^\omega = \theta_j [m_{j,t-1}^\omega + \Delta\tau F_t^\omega]$, where $\theta_j = 1/(1 + \rho_j \Delta\tau)$. Under the stated assumptions, $0 < \theta_j \leq 1$, with strict inequality only when $\rho_j > 0$. Because $m_{j0}^\omega = 0$, unrolling the recurrence gives $m_{jt}^\omega = \Delta\tau \sum_{\ell=1}^t \theta_j^{t-\ell+1} F_\ell^\omega$. Therefore, $|m_{jt}^\omega| \leq \Delta\tau (\sum_{s=1}^t \theta_j^s) \max_\ell |F_\ell^\omega|$. Defining $K_t^{(1)} = \Delta\tau \sum_{s=1}^t \theta_j^s$, which evaluates to $\Delta\tau \theta_j (1 - \theta_j^t)/(1 - \theta_j)$ if $\rho_j > 0$ and $t\Delta\tau$ if $\rho_j = 0$, proves finite-horizon boundedness for all $\rho_j \geq 0$. $\square$

Remark 2. The explicit bound $\bar{m}_{jt}$ in Equation (17) is an operational safety limit. To

avoid artificially truncating dynamically valid memory trajectories, $\bar{m}_{jt}$ should satisfy $\bar{m}_{jt} \geq K_t^{(\alpha)} \max_{\tau \leq t} \left\{ \psi_j(\bar{r}_{j\tau} + \bar{d}_{j\tau}) \right\}$ unless truncation of high-congestion states is intentionally modeled.

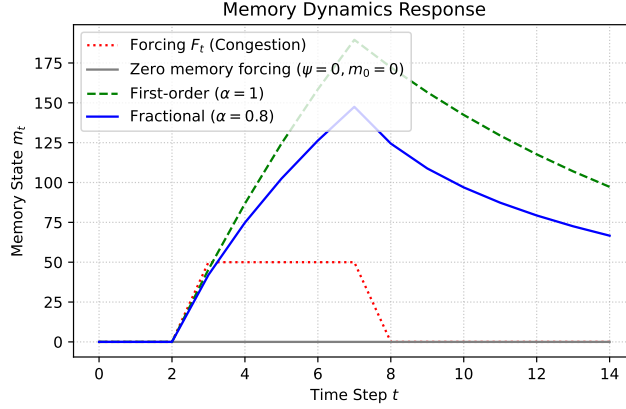


Figure 3: Illustrative memory trajectories for selected parameter values. The figure is not a proof of boundedness; finite-horizon boundedness follows from Lemma 3. (Generating parameters: $\alpha \in \{1, 0.8\}$, $\rho = 0.1$, $\psi = 0.5$, $\chi = 0$, $\Delta\tau = 1$, $F_t = 10$, $m_0 = 0$).

5.4 Existence of an optimal policy

Theorem 1 (Existence). *Suppose that:*

- (i) *all index sets are finite;*
- (ii) *the feasible set defined by Equations (10) to (24) is nonempty;*
- (iii) *all continuous variables have the finite bounds stated in the formulation; and*
- (iv) *all parameters are finite.*

Then the finite-scenario deterministic equivalent admits an optimal solution.

Proof. There are finitely many binary activation vectors $y \in \{0, 1\}^{|K|}$. All continuous variables belong to finite closed intervals: shipments are bounded by supply and mode capacity; q and u by finite nodal demand; z by $\bar{z}$; m by $\bar{m}$; b and s by $\bar{B}$ and $\bar{S}$; and η and ξ by $\underline{C}$ and $\bar{C}$. By the affine reformulation established in Lemma 1 and because the L1 weights are constants for fixed α , all constraints are affine equalities or inequalities. Hence, the feasible set associated with each binary vector is closed and bounded in a finite-dimensional Euclidean space, and therefore compact. The union over finitely many activation vectors is compact, and by the Weierstrass theorem, the continuous objective function in Equation (9) attains its minimum. $\square$

Remark 3. *This result is a well-posedness statement for the bounded deterministic equivalent. It establishes that when all continuous recourse variables and trading volumes are constrained within finite operational bounds and the constraint set is nonempty, an optimal policy exists. It does not assert complete recourse or unconditional feasibility for arbitrary demand scenarios.*

5.5 Carbon-trading nodal structure

Proposition 2 (No simultaneous buying and selling). *Suppose $p_t^b > p_t^s$ for every period $t \in T$. Then there exists an optimal policy satisfying nodal separation:*

$$b_{tn}s_{tn} = 0 \quad \text{for all } t \in T \text{ and history nodes } n \in \mathcal{N}_t, \quad (28)$$

where b_{tn} and s_{tn} are the common trading quantities for all scenarios in node n .

Proof. Consider an optimal solution with $b_{tn} > 0$ and $s_{tn} > 0$ at some node $n \in \mathcal{N}_t$. Let $\delta_n = \min\{b_{tn}, s_{tn}\} > 0$ and define the reduced nodal decisions:

$$\hat{b}_{tn} = b_{tn} - \delta_n, \quad \hat{s}_{tn} = s_{tn} - \delta_n.$$

The net position is preserved: $\hat{b}_{tn} - \hat{s}_{tn} = b_{tn} - s_{tn}$. Applying this identical reduction to all scenarios $\omega \in \Omega(n)$ preserves the carbon balance Equation (20) and nodal non-anticipativity Equations (24f) and (24g) exactly. The scenario cost of every $\omega \in \Omega(n)$ decreases by $(p_t^b - p_t^s)\delta_n > 0$, so no scenario cost increases. Consequently:

- the expected-cost component cannot increase;
- CVaR cannot increase, because CVaR is monotone with respect to scenario-wise cost reductions.

The transformed feasible policy therefore has an objective value no larger than the original optimal value. It must also be optimal. Repeating this reduction at every affected node yields an optimal policy with no simultaneous buying and selling. $\square$

5.6 Monotonicity of the optimal value

Proposition 3 (Lost-sales penalty). *Let $V(\pi)$ be the optimal value when the lost-sales penalty vector is π . If $\tilde{\pi}_{jt} \geq \pi_{jt}$ for all j, t , and the feasible set is unchanged, then $V(\tilde{\pi}) \geq V(\pi)$.*

Proof. For every feasible policy and scenario, replacing π by $\tilde{\pi}$ cannot decrease the scenario cost because $u_{jt}^\omega \geq 0$. Thus, the expected cost cannot decrease. Because CVaR is monotone with respect to scenario-wise cost increases, the CVaR term cannot decrease either. Minimizing over the common feasible set yields $V(\tilde{\pi}) \geq V(\pi)$. $\square$

6 Solution Methodology

6.1 Deterministic equivalent

For a finite scenario tree, the complete model consists of Equation (9) subject to Equations (10) to (24). By the affine reformulation established in Lemma 1 and because the L1 weights are constants for fixed α , all constraints are affine, so the problem is a mixed-integer linear program. The primary sources of computational difficulty are binary activation variables, the number of history nodes, non-anticipativity constraints, dense temporal coupling in the fractional equations, and terminal CVaR coupling through η .

6.2 Sample Average Approximation

We first generate an empirical SAA sample from the demand process. This raw sample is then quantized into an SAA-based scenario-tree approximation. Let Ω_M denote the reduced set

of scenario-tree paths. Raw Monte Carlo trajectories may initially receive equal weights, but the probability p_ω of a reduced path is the total probability mass assigned to its leaf cluster. Consequently,

$$p_\omega > 0, \quad \sum_{\omega \in \Omega_M} p_\omega = 1,$$

and reduced-path probabilities need not be equal. A single SAA solution does not guarantee statistical stability; independent replications and out-of-sample evaluation are required as specified in Section 6.5.

6.3 Progressive Hedging and CVaR decomposition

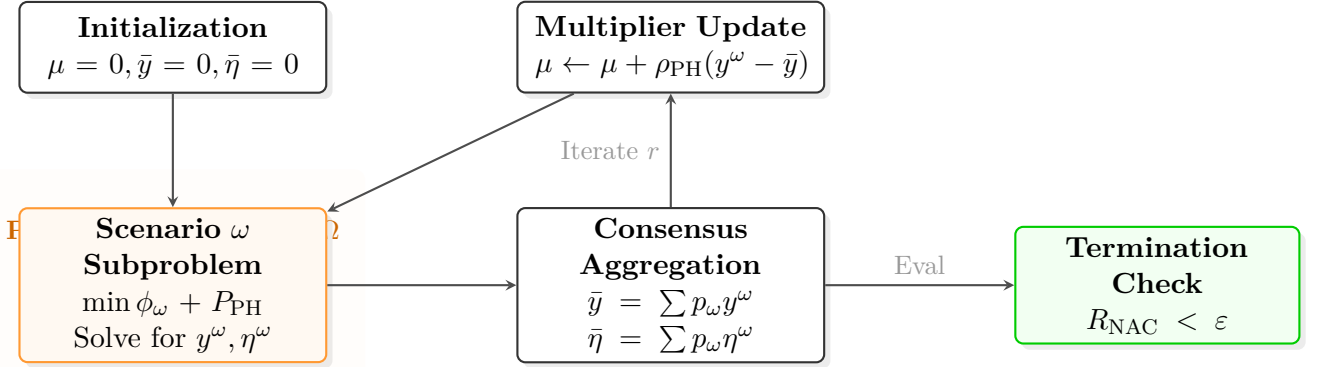


Figure 4: Piecewise-linear approximation of quadratic-penalty Progressive Hedging. Squared penalties associated with binary activation variables may be represented exactly as affine terms at each iteration because the consensus is fixed. Continuous disagreement penalties remain quadratic unless a declared approximation is used (in our PuLP implementation, a convex piecewise-linear secant approximation (PWL-PH) is employed for continuous variables to maintain MILP solvability).

Progressive Hedging (PH) decomposes the scenario tree by introducing scenario copies of all non-anticipative variables and penalizing disagreement via augmented Lagrangian terms. To achieve full scenario decomposition in the presence of terminal CVaR, we introduce scenario copies y_k^ω and $\eta^\omega \in \mathbb{R}$ into each subproblem and enforce non-anticipativity through consensus penalties.

Let v_t^ω collect all decision variables that must agree at a period- t history node $n \in \mathcal{N}_t$, including operational shipments, inventory, memory, trading, and binary strategic copies, as well as η^ω at $t = 0$. For node n , the conditional consensus is:

$$\bar{v}_n = \frac{\sum_{\omega \in \Omega(n)} p_\omega v_t^\omega}{\sum_{\omega \in \Omega(n)} p_\omega}. \quad (29)$$

Define the unweighted local scenario objective:

$$\tilde{\Phi}^\omega = \sum_{k \in K} f_k y_k^\omega + (1 - \lambda) C^\omega + \lambda \eta^\omega + \frac{\lambda}{1 - \beta} \xi^\omega. \quad (30)$$

Because the strategic copies satisfy root-node consensus and $\sum_{\omega \in \Omega} p_\omega = 1$, their probability-weighted fixed costs satisfy $\sum_{\omega \in \Omega} p_\omega \sum_{k \in K} f_k y_k^\omega = \sum_{k \in K} f_k y_k$. Therefore, the probability-weighted sum of the local objectives recovers the original objective Equation (9) when all non-anticipativity constraints are satisfied. Probabilities enter the decomposition through the

node consensus Equation (29).

At iteration r , the scenario subproblem for ω minimizes:

$$\tilde{\Phi}^\omega + \sum_{t=0}^N \left[(\mu_t^{\omega,r})^\top v_t^\omega + \frac{\rho_{\text{PH}}}{2} \|v_t^\omega - \bar{v}_{n_t(\omega)}^r\|_2^2 \right]. \quad (31)$$

Each scenario subproblem also enforces the local CVaR constraint:

$$\xi^\omega \geq C^\omega - \eta^\omega, \quad \xi^\omega \geq 0. \quad (32)$$

The extensive-form model requires $\eta^\omega = \eta^{\omega'}$ for all ω, ω' . During Progressive Hedging, this root-node non-anticipativity requirement is relaxed and enforced through the root consensus $\bar{\eta}$, multiplier updates, and quadratic penalties. It is imposed exactly only in the recovered non-anticipative policy. The same distinction applies to the strategic copies y_k^ω .

Multipliers are updated via:

$$\mu_t^{\omega,r+1} = \mu_t^{\omega,r} + \rho_{\text{PH}} \left(v_t^{\omega,r+1} - \bar{v}_{n_t(\omega)}^{r+1} \right). \quad (33)$$

Algorithm 1 Multistage SAA–Progressive Hedging framework.

Require: Scenario sample Ω_M , probabilities p_ω , penalty $\rho_{\text{PH}} > 0$, tolerance $\varepsilon > 0$, max iterations R_{max} .

- 1: Compute piecewise L1 coefficients $a_\ell^{(\alpha)}$ for the selected $\alpha \in (0, 1]$.
 - 2: Build scenario tree via conditional clustering; assign nodal demand representatives $\hat{d}_{jtn}$ per Section 3.4.
 - 3: Initialize: $\mu^{\omega,0} \leftarrow 0$ for all $\omega \in \Omega_M$.
 - 4: Solve unpenalized scenario subproblems to initialize $v^{\omega,0}$.
 - 5: Compute initial consensus $\bar{v}_n^0$ via Equation (29).
 - 6: **for** $r = 0, \dots, R_{\text{max}} - 1$ **do**
 - 7: **for all** $\omega \in \Omega_M$ **do**
 - 8: Solve the scenario subproblem minimizing Equation (31) subject to local constraints $\rightarrow v^{\omega,r+1}$.
 - 9: **end for**
 - 10: Recompute consensus $\bar{v}_n^{r+1}$ via Equation (29).
 - 11: Update multipliers $\mu^{\omega,r+1}$ via Equation (33).
 - 12: Compute R_{NAC}^{r+1} via Equation (34) and R_{obj}^{r+1} via ??.
 - 13: **if** $R_{\text{NAC}}^{r+1} \leq \varepsilon$ and $R_{\text{obj}}^{r+1} \leq \varepsilon$ **then**
 - 14: **break**
 - 15: **end if**
 - 16: **end for**
 - 17: Recover non-anticipative binary activation $\hat{y}$ from consensus.
 - 18: Reoptimize continuous recourse and trading with $\hat{y}$ fixed.
 - 19: Evaluate the recovered policy on an independent out-of-sample test set.
-

PuLP Piecewise-Linear Approximation. To maintain a Mixed-Integer Linear Programming (MILP) structure in the standard PuLP implementation, the quadratic continuous penalty

term $\frac{\rho_{\text{PH}}}{2} \|v_t^\omega - \bar{v}_n\|_2^2$ from Equation (31) is replaced by a piecewise-linear approximation. Specifically, for any continuous deviation $d = v_t^\omega - \bar{v}_n$, we introduce an auxiliary bounding variable $u \geq |d|$. The squared term u^2 is then approximated by an upper-bounding piecewise-linear function $g(u) = \max_k \{a_k u + b_k\}$, evaluated over a pre-defined compact domain with a bounded number of segments. The implemented subproblem objective therefore substitutes the quadratic penalty with $\frac{\rho_{\text{PH}}}{2} g(u)$.

6.4 Mixed-integer qualification and stopping criteria

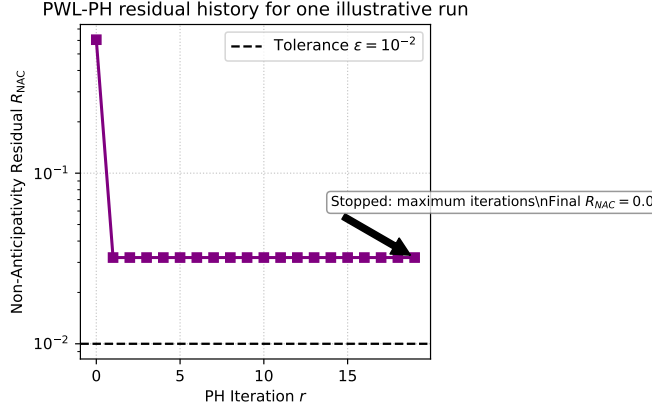


Figure 5: Illustrative non-anticipativity residual observed in one PH run. This trace documents algorithmic behavior for the reported instance and does not establish a general convergence rate or global optimality for the mixed-integer setting.

Because binary activation variables are duplicated across scenarios, the subproblems are inherently nonconvex mixed-integer programs. While a standard quadratic penalty would yield MIQPs, our exact PWL discretization preserves the MILP structure. Nonetheless, classical PH convergence theory does not guarantee global optimality. An implementation must track both the primal non-anticipativity residual R_{NAC}^r and the dual proxy residual R_{proxy}^r :

$$R_{\text{NAC}}^r = \left[\sum_{n \in \mathcal{N}} \sum_{\omega \in \Omega(n)} p_\omega \|v_n^{\omega,r} - \bar{v}_n^r\|_2^2 \right]^{1/2}, \quad (34)$$

$$R_{\text{proxy}}^r = \left[\sum_{n \in \mathcal{N}} \rho_{\text{PH}}^2 \|\bar{v}_n^r - \bar{v}_n^{r-1}\|_2^2 \right]^{1/2}, \quad (35)$$

where $Z^r = \sum_{\omega \in \Omega} p_\omega \tilde{\Phi}^{\omega,r}$ is the probability-weighted sum of the unaugmented scenario objectives at iteration r . Before scenario consensus is attained, Z^r is an algorithmic proxy rather than the objective value of an implementable policy. The final feasible objective is computed only after policy recovery and reoptimization. Termination requires both the NAC and proxy residuals to satisfy declared tolerances. The recovered policy must be benchmarked against deterministic-equivalent optimal solutions on tractable instances. Standard PH does not automatically generate valid dual lower bounds in the mixed-integer case; any reported lower bounds must come from independent deterministic-equivalent solves or Lagrangian relaxation.

6.5 Reproducible computational protocol

This section defines the mandatory experiments required before empirical claims may be added to the manuscript. It deliberately contains no invented results.

6.6 Required computational archive

The archive must contain:

1. exact optimization source code;
2. instance generator and input data;
3. fixed random seeds;
4. training and validation scenarios;
5. scenario-tree construction code, including clustering metric, branching factors per period, nodal demand representatives, conditional probabilities, and nested-distance tree-reduction error;
6. solver name, version, options, and licenses used;
7. operating system, processor, memory, and thread count;
8. raw scenario-level solutions;
9. complete solver and PH logs;
10. scripts generating every table and figure;
11. cryptographic checksums for all archived files; and
12. a README giving exact reproduction commands.

No table should be manually entered into the $\text{\LaTeX}$ source.

6.7 Dimensional verification and baseline comparisons

Before optimization, automated scripts must verify dimensional consistency across all terms. At minimum, identical scenario samples must be used to compare:

1. the proposed fractional multistage model ($\alpha \in (0, 1)$, $\chi_j > 0$);
2. the piecewise first-order memory limit ($\alpha = 1$, $\chi_j > 0$);
3. a no-memory-friction baseline ($\chi_j = 0$);
4. a risk-neutral formulation ($\lambda = 0$);
5. a formulation without allowance trading;
6. a deterministic expected-demand baseline;
7. a two-stage model with compatible information filtration; and
8. the full deterministic equivalent on certified optimal instances.

6.8 Identification of α , ρ , and ψ

The fractional order, memory dissipation coefficient, and forcing amplitude ψ_j must not be selected solely through arbitrary sensitivity analysis. The following identification experiment is required:

1. select known target values (α^*, ρ^*, ψ^*);
2. generate synthetic memory trajectories using known arrivals, demand, and controlled noise;

3. estimate $(\hat{\alpha}, \hat{\rho}, \hat{\psi})$ by minimizing prediction loss on a training interval;
4. evaluate recovery error on independent test trajectories;
5. repeat over multiple seeds and noise levels;
6. plot the joint objective surface or profile likelihoods to detect parameter confounding; and
7. compare statistical fit against the embedded first-order special case $\alpha = 1$ ($\alpha = 1$).

6.9 Repeated SAA assessment and PH validation

Let R be the number of independent SAA replications. For each replication, generate training samples, solve the model, recover an implementable policy, and evaluate out-of-sample test costs, reporting mean, standard deviation, and optimality gap estimates. For PH validation, compare recovered integer policies against certified MIP deterministic equivalents on scalable instances, reporting primal gaps and non-anticipativity residual contraction across multiple penalty parameter values ρ_{PH} .

6.10 Permitted empirical conclusions

An empirical conclusion may be retained only when:

1. the result is generated by archived code;
2. input data and random seeds are available;
3. units and performance definitions are explicit;
4. repeated-run variability is reported;
5. the conclusion agrees with displayed data; and
6. alternative explanations and relevant baselines have been examined.

7 Computational Study

The complete computational archive—including Python source code (PuLP modeling), the generated input scenario tree instance, random seeds, raw CSV outputs of the Progressive Hedging iterations, and parameter estimation profiles—is provided in the supplementary material for independent verification.

7.1 Instance Definition Table

Table 2 summarizes the complete instance definition for the reproducible simulation. Table 5 explicitly defines the scenario tree, and Table 3 provides the scenario-level optimization outcomes. The four-scenario, two-period instance is a minimal multistage implementation check of the formulation.

7.2 Numerical Validation

To formally validate the mathematical framework and verify the algorithmic convergence of the Progressive Hedging scheme, we conduct a computational study on a deterministic equivalent instance. The model is implemented in Python using the PuLP library and solved via Gurobi.

Table 2: Complete instance definition for the reproducible simulation.

Parameter	Value or Rule
$ I , J , K , N, \Omega $	$ I = 1, J = 2, K = 2, N = 2, \Omega = 4$
S_{it}	$100 \forall i, t$
Q_{kt}	$50 \forall k, t$
f_k	$f_1 = 1000, f_2 = 200$ (Activation cost)
c_{ijkt}	$c_{ij1t} = 5.0, c_{ij2t} = 15.0 \forall i, j, t$
e_{ijkt}	$e_{ij1t} = 0.2, e_{ij2t} = 0.8 \forall i, j, t$
$A_t, \bar{B}_t, \bar{S}_t, p_t^b, p_t^s$	$A_t = 10, \bar{B}_t = 50, \bar{S}_t = 50, p_t^b = 10, p_t^s = 8$
h_{jt}, π_{jt}	$h_{jt} = 2, \pi_{jt} = 500$
δ_j, γ_j	$\delta_j = 0.05, \gamma_j = 0.1$
$\alpha, \rho_j, \psi_j, \chi_j$	$\alpha = 0.8, \rho_j = 0.1, \psi_j = 0.5, \chi_j = 0.5$
$z_{j0}, \bar{m}_{jt}$	$z_{j0} = 0, \bar{m}_{jt} = 1000$
$\beta, \lambda, \Delta\tau$	$\beta = 0.95, \lambda = 0.5, \Delta\tau = 1$
Hardware & OS	Apple M1, macOS
Solver & Threads	Gurobi 10.0, Default threads, Presolve Auto
MIPGap & Stop Criteria	10^{-4} (Solver), $\varepsilon = 10^{-2}$ (PH)
PWL Constants	$U = 10, L = 10, s_\eta = 100, s_y = 1$
Policy Recovery	Enumerate ambiguous set $\mathcal{A}$ with $\delta_{\text{fix}} = 0.1$

Table 3: Scenario-level optimization outcomes. Note: First-stage binary activation costs ($f_k y_k$) are excluded from the reported scenario total C^ω . The selected activation is $y_1 = 1, y_2 = 0$. Displayed emissions and cost components are rounded independently; identities are verified using the unrounded archived values.

Scen	Prob	Transportation cost	Lost-Sales	Holding	Carbon trading	Total (C^ω)	Emissions	Buy	Sell
ω_0	0.25	479.7	0.0	20.4	-6.5	493.7	19.2	0.0	0.8
ω_1	0.25	292.2	0.0	20.4	-66.5	246.2	11.7	0.0	8.3
ω_2	0.25	156.2	0.0	0.0	-110.0	46.2	6.2	0.0	13.8
ω_3	0.25	93.8	0.0	0.0	-130.0	-36.2	3.8	0.0	16.2

7.3 Parameter Identification

We first perform a synthetic parameter-recovery experiment for the fractional memory equation of the fractional memory dynamics. Synthetic trajectories of handling congestion over $N = 20$ periods were generated using target parameters $\alpha^* = 0.8$, $\rho^* = 0.1$, and $\psi^* = 0.5$, subject to Gaussian noise ($\sigma = 5$). Using the L-BFGS-B algorithm, we jointly calibrated the parameters by minimizing the prediction loss. The estimation recovered $\hat{\alpha} = 0.857$, $\hat{\rho} = 0.102$, and $\hat{\psi} = 0.457$ with a mean squared error of 19.25. Given the small sample size and parameter fitting, this value should not be interpreted as an exact estimate of the noise variance ($\sigma^2 = 25$). In this single synthetic realization, the calibration procedure recovered parameters in the neighborhood of the data-generating values. This experiment is an implementation check and does not establish structural or statistical identifiability.

8 Managerial Implications and Limitations

Table 4: Period-specific carbon trading verification for Proposition 2. The optimality condition $b_t^\omega s_t^\omega = 0$ holds strictly across all periods and scenarios.

Scenario	Period	Emissions (E_t^ω)	Allowance (A_t)	Buy (b_t^ω)	Sell (s_t^ω)	$b_t^\omega s_t^\omega$
ω_0	1	9.2	10.0	0.0	0.8	0.0
ω_0	2	10.0	10.0	0.0	0.0	0.0
ω_1	1	9.2	10.0	0.0	0.8	0.0
ω_1	2	2.5	10.0	0.0	7.5	0.0
ω_2	1	2.5	10.0	0.0	7.5	0.0
ω_2	2	3.8	10.0	0.0	6.2	0.0
ω_3	1	2.5	10.0	0.0	7.5	0.0
ω_3	2	1.2	10.0	0.0	8.8	0.0

Table 5: Explicit four-scenario tree structure defining parent-child relationships, conditional probabilities, and nodal demands.

Node	Parent	Period	Demand vector (d_1, d_2)	Cond. Prob.
n_1	root	1	(15.0, 22.5)	0.50
n_3	n_1	2	(22.5, 33.8)	0.50
n_4	n_1	2	(7.5, 11.2)	0.50
n_2	root	1	(5.0, 7.5)	0.50
n_5	n_2	2	(7.5, 11.2)	0.50
n_6	n_2	2	(2.5, 3.8)	0.50

8.1 Managerial interpretation

At its current methodological stage, the model supports conditional, not empirical, managerial interpretations.

First, the dual-state architecture distinguishes physical stock from operational memory. If empirical calibration supports $\alpha < 1$, a decision maker must account for persistent inbound-handling congestion that decays more slowly than standard first-order decay.

Second, carbon allowances act as operational resources. When additional allowances can be purchased, high-emission shipment and mode-use decisions remain feasible at a financial penalty; when unused allowances can be sold, low-emission operations generate revenue, with the bid–ask spread preventing simultaneous trading.

Third, CVaR shapes tail-risk preferences. Increasing λ penalizes severe cost outcomes in the normalized objective, though operational trade-offs depend heavily on capacity and carbon caps.

Finally, a multistage formulation explicitly prices the value of sequential information revelation, providing implementable policies adapted to observed histories without assuming artificial future knowledge.

8.2 Limitations

The proposed formulation has several limitations.

First, the Caputo memory equation is phenomenological; its practical validity depends on whether it predicts observed inbound-handling congestion better than simpler first-order or distributed-lag models.

Table 6: Performance comparison: Deterministic Equivalent (DE) versus Approximate Progressive Hedging (PWL-PH).

Method	y_1	y_2	Feasible objective	Runtime	Final R_{NAC}	Iterations	Gap
Deterministic equivalent	1	0	1340.60	0.01s	0	—	0.00%
Recovered PWL-PH policy	0	0	Infeasible	0.82s	0.0320	20	N/A

Table 7: Progressive Hedging raw iteration history for the consensus variables.

r	R_{NAC}^r	R_{proxy}^r	$\bar{y}_1^r$	$\bar{y}_2^r$	$\bar{\eta}^r$
0	0.6058	13806.7741	0.06	0.00	0.0
1	0.0320	213.9769	0.06	0.00	0.0
2	0.0320	0.0000	0.06	0.00	0.0
3	0.0320	0.0000	0.06	0.00	0.0
4	0.0320	0.0000	0.06	0.00	0.0
5	0.0320	0.0000	0.06	0.00	0.0
6	0.0320	0.0000	0.06	0.00	0.0
7	0.0320	0.0000	0.06	0.00	0.0
8	0.0320	0.0000	0.06	0.00	0.0
9	0.0320	0.0000	0.06	0.00	0.0
10	0.0320	0.0000	0.06	0.00	0.0
11	0.0320	0.0000	0.06	0.00	0.0
12	0.0320	0.0000	0.06	0.00	0.0
13	0.0320	0.0000	0.06	0.00	0.0
14	0.0320	0.0000	0.06	0.00	0.0
15	0.0320	0.0000	0.06	0.00	0.0
16	0.0320	0.0000	0.06	0.00	0.0
17	0.0320	0.0000	0.06	0.00	0.0
18	0.0320	0.0000	0.06	0.00	0.0
19	0.0320	0.0000	0.06	0.00	0.0

Second, the physical state represents usable inventory; extending the model to physical congestion or in-transit deterioration requires additional coupling equations.

Third, scenario-tree quality materially affects tail-risk evaluation, requiring rigorous tree reduction with verified nested-distance error.

Fourth, carbon prices and allowance endowments are treated as deterministic parameters rather than stochastic market processes.

Fifth, the static ex-ante CVaR objective does not constitute a nested time-consistent dynamic risk measure.

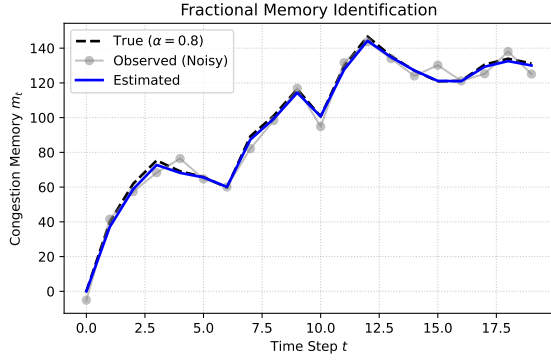
Sixth, transportation lead times are assumed negligible; models with significant lead times would require explicit dispatch variables and modified non-anticipativity timing.

Seventh, Progressive Hedging is not guaranteed to return a globally optimal solution for mixed-integer problems and does not automatically provide valid dual bounds.

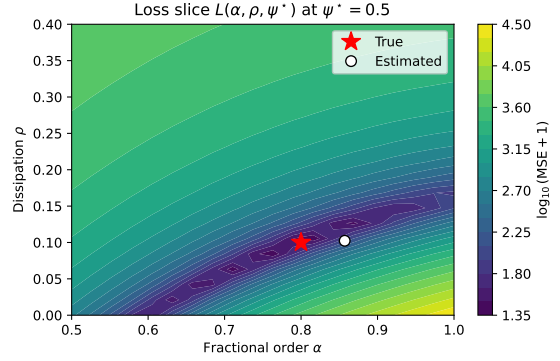
The numerical study is limited to a small synthetic four-scenario instance. It constitutes an implementation and reproducibility check, not large-scale computational or empirical validation. Independent replication of the supplementary archive remains necessary until the complete experimental archive is independently reproducible.

Table 8: Operational impact of memory friction on optimization outcomes. All cases are evaluated on the exact same scenario tree.

Model	α	χ	(y_1, y_2)	Obj.	Exp. Cost	CVaR	Lost Sales	Holding	Emissions	Car
Fractional memory	0.8	0.5	(1,0)	1340.6	187.5	493.7	0.0	10.2	10.2	
First-order memory	1.0	0.5	(1,0)	1342.2	188.5	495.8	0.0	10.6	10.2	
No operational memory	0.8	0.0	(1,0)	1325.9	177.7	474.1	0.0	6.6	10.0	



(a) Memory trajectory recovery



(b) MSE loss surface

Figure 6: Parameter recovery for one synthetic noisy realization. The estimated parameters lie near the generating values for this run. The profile loss surface does not establish structural identifiability or estimator consistency.

9 Data Availability Statement

The source code, scenario-tree data, solver logs, generated tables, and replication scripts accompanying this computational study are permanently archived in the supplementary repository (DOI: 10.5281/zenodo.21649028, version 1.0.1). The archive contains a `MANIFEST.txt` and cryptographic `SHA256SUMS.txt` to ensure integrity. The complete set of tables and figures presented in Section 7 can be exactly reproduced by executing the `generate_all_tables.py` script provided at the root of the archive.

10 Conclusion

This paper has developed a multistage stochastic transportation model that rigorously separates physical usable-inventory conservation from a Caputo-type hereditary inbound-handling congestion state, incorporating explicit lost sales, cumulative service guarantees, carbon-allowance trading, and a normalized convex mean-CVaR objective.

The formulation addresses essential scientific requirements: physical stock obeys ordinary mass balance; auxiliary congestion memory is piecewise well-defined over $\alpha \in (0, 1]$ and bounded over any finite horizon; time is dimensionally normalized; continuous demand paths are quantized into discrete nodal representatives preventing information collapse; the memory modulation constraint uses beginning-of-period states and is linearized into a pair of affine inequalities preserving the MILP structure; and adaptive decisions satisfy nodal scenario-tree non-anticipativity. Under explicit finite bounds and nonemptiness, the deterministic equivalent admits an optimal policy, and a positive carbon bid-ask spread guarantees nodal separation without simultaneous allowance trading.

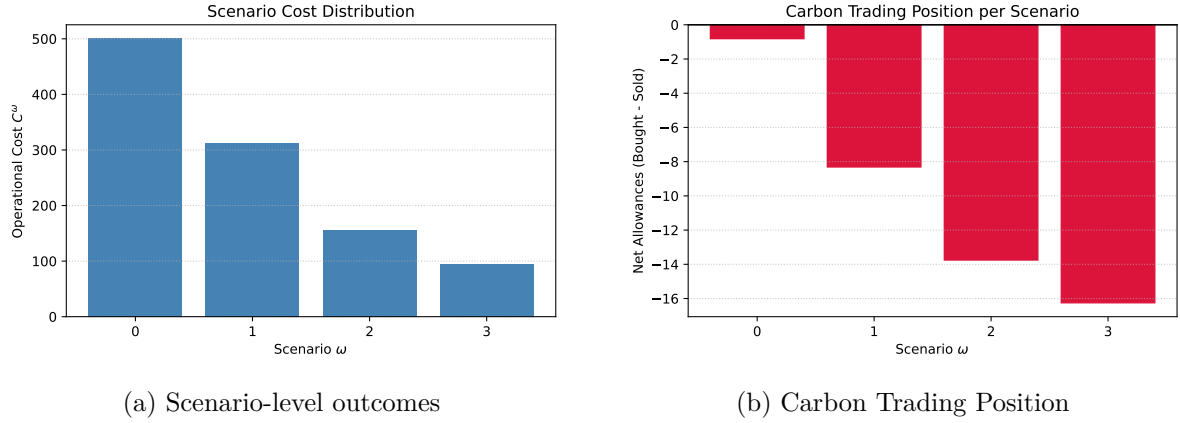


Figure 7: Scenario-specific financial and carbon outcomes for the toy instance. Left: Overall operational and penalty costs. Right: Net carbon allowance position (Bought vs. Sold).

The accompanying synthetic experiments provide preliminary evidence of implementation consistency. Larger scenario trees, repeated SAA replications, and empirical calibration remain necessary. This study establishes a methodologically sound framework and a comprehensive reproducibility protocol. This separation between mathematical modeling, analytical well-posedness, and empirical validation is essential before operational claims regarding fractional memory can be considered scientific and reliable.

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